

Understanding Overspending: An Explainable AI Approach to Financial Behavior

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Abstract

This project introduces a personal financial risk explainable machine learning framework (XAI) that provides end users with predictive performance and interpretability. Previous financial risk detection systems act as black boxes and often limit a user’s ability to understand why certain transactions of theirs are labeled as risky. To address this common issue, we created a pipeline that combines behavioral feature engineering with XAI to give users transparent insights. We evaluated three models based on standard evaluation metrics: logistic regression, random forest, and gradient boosting. We developed a UI/UX that enables users to input transaction data and receive a fraud prediction, along with a clear explanation of the model’s decision. Our system shows that explainability and a straightforward UI/UX feature makes personal financial risk detection accessible to everyday people, offering a heightened transparency that is not typically seen in the data science realm.

Keywords

overspending prediction, explainable AI, machine learning, behavioral finance, SHAP

1 Introduction

Personal financial instability can often be predicted long before major financial pitfalls like low credit scores and loan delinquency. Subtle shifts in daily spending can be used to signal emerging risk long before static financial metrics detect it. By using machine learning, we can learn a user’s typical monthly spending habits, which can help detect an out of character transaction and decide whether or not it should be labeled as risky activity.

This project develops an early warning system that is designed to detect individual financial risks in an accessible and interpretable way. We built a personal financial risk system that incorporates explainable AI (XAI) to help users understand not just whether a transaction was flagged, but why it was. Our goal is to change static, simple predictions into personalized, meaningful insight.

To support this goal, we designed a UI/UX that allows users to input time aware financial indicators, including monthly income, category spending (e.g., entertainment, health and fitness, shopping, utilities, food and drink, rent, travel), and prior spending history. These inputs capture gradual behavioral patterns that traditional

risk models often overlook. The system then generates an overspending risk detection report that indicates whether a transaction was flagged and provides an explanation for the model’s decision.

By combining behavioral feature engineering, machine learning, and XAI, this work aims to transform fraud detection tools from static black boxes into transparent, user-centered systems that make personal financial risk easier to understand.

2 Literature Review

The intersection of machine learning, behavioral financial indicators, and explainable AI forms the theoretical and methodological foundation of this project. This review synthesizes five prior works that collectively justify the shift from institutional credit-risk modeling to individual-level overspending forecasting and establish the case for transparency through Explainable AI (XAI).

2.1 Machine Learning in Financial Forecasting

The application of machine learning to financial prediction has expanded substantially over the past decade. Khattak et al. conducted a systematic review of 51 studies spanning 2018 to 2023, examining AI techniques across stock, cryptocurrency, forex, and commodity markets [3]. Their findings confirm that hybrid and ensemble models—particularly LSTM-based architectures combined with SVM, MLP, and CNN—consistently outperform standalone approaches in predictive performance. Despite these advances, the review identifies a critical gap: most studies optimize for performance metrics such as RMSE and MAPE while neglecting profitability evaluation, real-world trading validation, and multi-output prediction frameworks. This observation motivates the present project’s emphasis on system-level behavioral modeling rather than single-metric optimization, and underscores the need for approaches that extend beyond narrow forecasting accuracy toward practical, interpretable financial utility.

2.2 From Institutional Credit Risk to Individual Behavioral Risk

Traditional financial risk models have been designed predominantly from an institutional perspective. Khandani, Kim, and Lo demonstrate that nonlinear machine learning models applied to consumer banking transaction data significantly outperform conventional

credit scoring methods in forecasting delinquencies [2]. By incorporating behavioral indicators—including spending patterns, income fluctuations, and debt ratios—their models detect early warning signals of financial distress with greater sensitivity than static scoring systems. While this work establishes the predictive value of transaction-derived behavioral data, it remains oriented toward lender risk management rather than individual financial awareness. This project builds directly on their behavioral insight but reorients the modeling objective: rather than forecasting delinquency for institutional use, we forecast next-month overspending risk at the personal level using the same class of transaction-derived behavioral signals.

2.3 Forecasting vs. Planning: A Critical Methodological Distinction

A key methodological boundary addressed by this project is the distinction between forecasting and causal planning. Wasserbacher and Spindler examine this boundary directly, demonstrating through both literature review and simulation experiments that while machine learning models such as LASSO substantially improve out-of-sample forecasting accuracy, the naive application of predictive models to planning or intervention problems produces biased and misleading conclusions [4]. Consistent with their argument, our project positions itself exclusively as a forecasting and behavioral transparency tool rather than a causal intervention system. Feature-level contributions derived from SHAP and LIME are interpreted as statistical associations within the forecasting structure, not as causal levers for financial decision-making, which preserves the integrity of model outputs and avoids overstating the prescriptive capacity of the system.

2.4 Explainability as a Requirement in Financial AI

As AI models grow more complex, the inability to interpret their outputs becomes a significant barrier to trust, adoption, and regulatory compliance—particularly in financial contexts. Yeo et al. provide a comprehensive review of XAI methods in finance, introducing the concept of FinXAI and distinguishing between model explainability (understanding how a model operates globally) and outcome explainability (understanding why a specific prediction was made) [5]. The authors categorize relevant XAI techniques including feature importance methods, surrogate models such as LIME, and counterfactual explanations, while emphasizing the fundamental tradeoff between accuracy and interpretability. Crucially, they argue that explanations must be tailored to specific audiences: end-users require simple local explanations about their individual risk, while regulators and developers need global model-level transparency. This audience-centered framework directly informs our integration of SHAP and LIME, which will decompose predictions into feature-level behavioral contributions presented in terms accessible to individual users.

2.5 Conversational AI as a Transparency Delivery Layer

Predictive accuracy and interpretability are necessary but insufficient if model outputs cannot be communicated effectively to end users. Kamra et al. present an expense monitoring system that integrates machine learning-based transaction categorization with a conversational AI interface, enabling users to query financial summaries and trend analyses through natural language [1]. The system demonstrates that complex financial data can be translated into intuitive, user-facing insights without requiring technical literacy, achieving transaction categorization accuracy above 90%. While their framework does not address the logic underlying the predictions it surfaces, it provides a practical blueprint for how conversational AI can serve as a transparency delivery layer—communicating behavioral risk signals in plain, actionable terms. This aligns with the output design objectives of our project, where SHAP- and LIME-derived behavioral contributions are intended not just for model evaluation but for meaningful user-facing early-warning communication.

2.6 Synthesis and Gap

Taken together, these works establish three converging pillars for this project. First, machine learning models leveraging behavioral transaction data demonstrably outperform static financial ratios in predicting financial risk. Second, the responsible application of these models requires a clear methodological boundary between forecasting and causal planning. Third, model outputs must be explainable and audience-appropriate to generate trust and actionable insight. The gap that remains—and that this project directly addresses—is the absence of a unified behavioral forecasting pipeline that combines transaction-level feature engineering, supervised learning, and XAI decomposition oriented specifically toward individual-level overspending prediction rather than institutional credit evaluation.

3 Methodology

3.1 Dataset and Preprocessing

3.1.1 Dataset Selection. This project underwent an important dataset transition during development. Initially, the experiments were trained and evaluated on a smaller curated dataset with raw credit card transaction records which were used to construct monthly behavioral summaries. Upon performing experiments on this dataset, it became clear there were too few cases for any of the three models to learn from. The team responded by transitioning to the Sparkov synthetic credit card transactions dataset, sourced from Kaggle. Sparkov dataset contains over 1.8 million transaction records for over 1,000 customers in a two-year period. Relevant features include personally identifiable information (PII) such as name, transaction date, spending category, and transaction amount.

3.1.2 Preprocessing Steps. The Sparkov dataset underwent several preprocessing steps to adapt from a credit card fraud detection dataset to a personal finance forecasting dataset, including aggregation, income mapping, and cleansing. The first step was to consolidate raw credit card transactions to create profiles stratified by individual and by month. This was done by grouping transactions

by the first and last name, month and year, and purchase category attached to that transaction. The individual's occupation was also used as a grouping condition to mitigate duplicate names in the dataset. Now that individuals' monthly purchase history by category was aggregated, a pivot table was used to aggregate category totals by person and month/year into a single row. Once consolidated, rows containing any missing (NaN) values for any category were discarded.

One of the advantages of the originally-used dataset is that despite the lack of cases, there were reliable figures for measuring individuals' income, a vital metric when considering overspending. While no such variable existed in this dataset, the job column offers a proxy. To capture an estimated income for each individual, we fed the list of occupations in the dataset into Claude AI, which outputted the median salary of each. This median salary was divided by twelve to use as a monthly estimated income. The final filtered, preprocessed dataset resulted in 7,297 individual-monthly spending profiles for training, validation and testing.

3.2 Feature Engineering

The project experiments with a behavioral engineering framework that transforms raw monthly spending data into behavioral indicators. This novel approach attempts to move beyond static financial ratios to better learn the temporal and habitual patterns of individuals' personal finances. Appended features span four categories: Trends and volatility, frequency and regularity, habit-based indicators, and target variable construction (ground truth).

Trends and volatility feature engineering captured temporal spending dynamics. One such added variable, the rolling 3-month average (rolling_3m_expense) establishes a stable behavioral baseline by smoothing short term fluctuations. Expense_mom_change is a variable that quantifies the rate at which spending accelerates or decelerates, thus serving as an early warning signal.

The frequency and regularity feature engineering steps attempt to capture how frequently an individual transacts and whether those patterns are stable over time. Features include transaction count, average inter-transaction gap, and monthly spending dispersion. Through this, large spending bursts are captured to better measure behavioral instability.

Habit-based indicators better illuminate patterns such as the ratios of discretionary spending and savings, as well as other flags such as overspending history and accounting for recurring expenses. Putting greater emphasis on these metrics is important because excess discretionary spending is a leading behavioral indicator of unsustainable habits.

Lastly, a target, or ground truth, variable was added by calculating an overspending threshold for individuals' each month. The threshold was defined as 80% of income to maintain a savings buffer in accordance with the 50/30/20 principle.

3.3 Modeling

3.3.1 Experimental Setup. We evaluate three different classifiers, which are Logistic Regression (LR), Random Forest (FR), and Gradient Boosting (GB) to predict next-month overspending using the Sparkov dataset. Gradient boosting is a powerful machine learning approach that seeks to learn iteratively off of training data by

correcting mistakes made in its previous iteration. Random forest learning involves creating multiple decision trees simultaneously based off subsets of the training dataset. These multiple trees are then all used on the same test data, and use a voting system to determine how to classify target data. Logistic regression is a classification algorithm that models the probability of a binary outcome by fitting a linear combination of input features to a sigmoid function that outputs values between 0 and 1. Despite its name containing "regression," it is used for classification tasks and its strength lies in interpretability, as each feature has a coefficient that directly indicates how much it increases or decreases the predicted probability of overspending. After preprocessing and feature engineering, the modelling table contains 6,366 customer-month observations with a class balance of 58.8% non-overspend and 41.2% overspend in the following month. To evaluate the model's performance, two approaches are used: a time-aware hold-out split and a 4-fold expanding window walk-forward split. The time-aware hold-out split is where the most recent portion of the data is used as a test set and the rest of the earliest data is used for training. The walk-forward evaluation split is an approach where the model is trained multiple times on progressively larger chunks of past data and tested on the next time period. This process is repeated four times, ensuring that the model is always evaluated on data that comes after the training data, which better reflects real-world prediction scenarios. We also use evaluation metrics such as F1, ROC-AUC, precision, recall, and average precision (AP) to evaluate the models' performance. To address the class imbalance, we added two weight settings: default 0:1, 1:2 and a milder 0:1, 1:1.5. Optuna-based hyperparameter tuning (50 trials) is also applied to all three models.

Hyperparameter tuning was conducted on all three models to determine the best specifications for training. The following results were produced: gradient boosting (n_estimators = 190, learning rate = 0.084, max_depth = 2), random forest (337 trees, max_depth = 8), logistic regression (C = 0.030).

4 Result

4.1 Cross-Validated Performance

Table 1 summarizes walk-forward performance across folds for both the engineered feature set and the raw-feature ablation. Through implementation of engineered features and default class weights, Gradient Boosting produces the highest mean F1 (0.767) and the strongest precision (0.712), while Logistic Regression produces the highest recall (0.940) and average precision (0.830). All three models performed relatively well for AUC-ROC (0.894-0.895). As for the milder class weight ratio, there are slight improvements for LR and RF (F1 +0.006 each) and this does not affect GB. This means that the default weight already over-emphasized the minority class for boosted trees.

Table 1: Mean Walk-Forward Results Across Feature Sets and Class-Weight Settings

Setting	Model	F1	AUC	Prec.	Rec.	AP
Eng (default)	LR	0.755	0.894	0.665	0.940	0.830
Eng (default)	RF	0.761	0.895	0.690	0.918	0.827
Eng (default)	GB	0.767	0.894	0.712	0.899	0.814
Eng (mild)	LR	0.761	0.895	0.681	0.927	0.830
Eng (mild)	RF	0.767	0.896	0.709	0.903	0.824
Eng (mild)	GB	0.767	0.894	0.712	0.899	0.814
Raw (abl)	LR	0.755	0.892	0.677	0.922	0.823
Raw (abl)	RF	0.749	0.886	0.683	0.903	0.818
Raw (abl)	GB	0.748	0.887	0.730	0.838	0.816

4.2 Feature Engineering Ablation

To assess the impact of feature engineering, 50+ engineered features (trends, volatility, frequency/regularity, habit indicators) are removed, while raw inputs such as monthly category spending and demographic information are retained. The results show that there is only a small decrease in performance. For example, Table 2 shows that the F1 score for GB drops from (0.767) to (0.748), while LR remains unchanged (0.755 $\rightarrow$ 0.755). ROC-AUC score on the other hand decreases by less than 0.01 for all models. Overall, while feature engineering only provides a slight improvement, this improvement can be helpful for tree-based models.

Table 2: Mean Walk-Forward Results – Feature Engineering Ablation

Model	F1	AUC	Prec.	Rec.	AP
LR	0.755	0.892	0.677	0.922	0.823
RF	0.749	0.886	0.683	0.903	0.818
GB	0.748	0.887	0.730	0.838	0.816

4.3 Performance Under Temporal Distribution Shift

Per-fold inspection (Table 1) reveals that performance varies significantly over time. For example, Folds 1–3 produce strong and stable performance for GB (F1 = 0.84–0.90, ROC-AUC $\approx$ 0.93–0.96), but fold 4 performance scores drop to F1 = 0.47 and precision = 0.31, with ROC-AUC dropping to 0.76. This drop might be because of the proportion of overspending cases in the test set that decreases from 0.56 in fold 3 to 0.22 in fold 4. This means that the model is tested on data with a much lower rate of overspending than what it was trained on.

4.4 Hyperparameter Tuning

Optuna tuning on the hold-out split use shallow gradient boosting (n_estimators = 190, learning rate = 0.084, max_depth = 2), a moderately regularized random forest (337 trees, max_depth = 8), and a strongly regularized logistic regression (C = 0.030). However, as shown in (Table 3), the performance on the hold-out set is lower (F1 around 0.47) because this test set reflects the same distribution shift seen in fold 4. Overall, compared to previous results, hyperparameter tuning produces only minor improvements which suggests

that model performance is primarily limited by the data rather than the choice of hyperparameters.

Table 3: Performance After Hyperparameter Tuning (Hold-Out Set)

Model	Acc.	F1	Prec.	Rec.	AUC	AP
RF	0.553	0.474	0.318	0.935	0.770	0.464
GB	0.559	0.474	0.319	0.920	0.753	0.420
LR	0.552	0.472	0.316	0.927	0.763	0.500

4.5 Feature Attribution

Figure 1 showed TreeSHAP values computed on the tuned GB, which identify monthly savings, savings rate, and median monthly income as the top predictors.

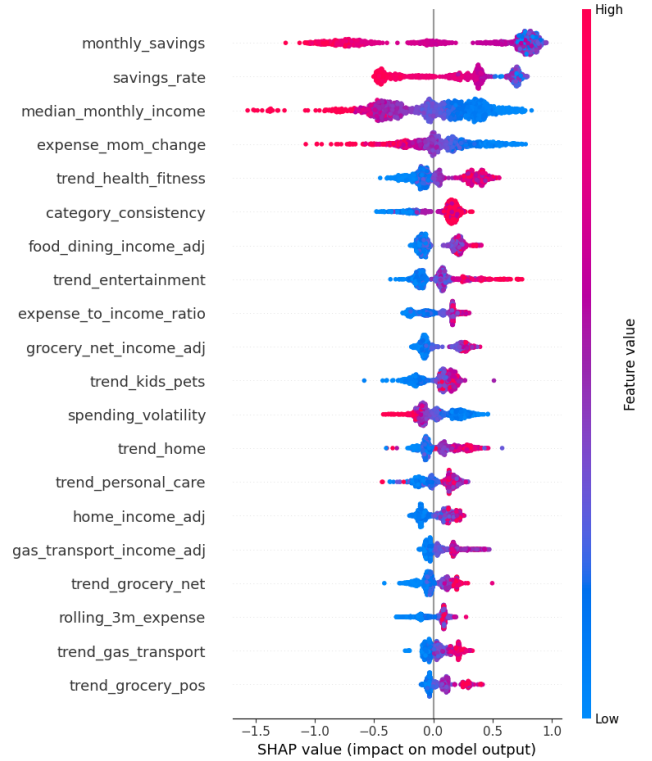


Figure 1: SHAP summary plot showing feature importance and their impact on model predictions. Higher SHAP values indicate a greater contribution toward predicting overspending.

5 Discussion

The results are consistently showing Gradient Boosting and Random Forest to be performing well in precision-focused performance for the walk-forward evaluation, while Logistic Regression demonstrates stronger recall and average precision. This divergence reflects a fundamental tradeoff in overspending prediction where precision-oriented models minimize false alarms and recall-oriented

models prioritize capturing as many true overspending cases as possible. In a personal finance early-warning context, higher recall is arguably more desirable since failing to warn a user who will overspend carries greater practical cost than issuing a cautionary signal that proves unnecessary.

5.1 Cross-Validated Model Behavior

Across all three classifiers, ROC-AUC scores remained competitive and closely clustered between 0.894-0.896 which suggests that the behavioral feature set provides a consistent and learnable signal regardless of model architecture. Gradient Boosting produced the highest mean F1 score of 0.767 and a precision of 0.712 under default class weights, while Logistic Regression achieved the strongest recall of 0.940 and average precision of 0.830. The fact that a regularized linear model performs comparably to ensemble methods implies that the predictive relationship between behavioral features and overspending risk is largely linear in nature, which has favorable implications for deployment. Overall, Logistic Regression offers lower computational overhead, faster inference, and more straightforward global interpretability than tree-based models.

5.2 Marginal Value of Feature Engineering

The feature engineering reveals that removing over 50 behavioral indicators which includes spending volatility, trend features, frequency and regularity measures, and habit signals, results in only modest performance degradation. The F1 score for Gradient Boosting declined from 0.767 to 0.748, Logistic Regression remained unchanged at 0.755, and ROC-AUC decreased by less than 0.01 across all models. This suggests that raw monthly category spending and demographic features already encode a substantial share of the predictive signal. Engineered features provide incremental refinement especially for tree-based models. This finding does not diminish the value of behavioral feature engineering but rather it indicates that income and savings structure constitute the dominant predictive signal and that higher-order behavioral volatility measures may contribute more meaningfully over longer training horizons or with more demographically diverse populations.

5.3 Temporal Distribution Shift

The most consequential finding is the sharp degradation in fold 4 of the walk-forward evaluation. Gradient Boosting F1 score fell from a stable range of 0.84-0.90 in folds 1-3 to 0.47 in fold 4, with precision dropping to 0.31 and ROC-AUC declining to 0.76. This deterioration coincides with a substantial reduction in the proportion of overspending cases in the 4 fold test set, from 0.56 in fold 3 to 0.22 in fold 4, which indicates that all three models encountered a class distribution markedly different from their training data. The hold-out evaluation results, which yield F1 scores near 0.47 across all models, confirm that this is a systematic issue rooted in distributional drift rather than any modeling deficiency. Optuna-based hyperparameter tuning across 50 trials produced only marginal improvements, further supporting the conclusion that performance is primarily bounded by data distribution rather than suboptimal configuration.

5.4 Explainability and Behavioral Transparency

TreeSHAP analysis on the tuned Gradient Boosting model identifies monthly savings, savings rate, and median monthly income as the top predictors of next-month overspending risk. This result is behaviorally coherent since individuals with compressed savings buffers and narrow income-to-expense ratios are structurally more vulnerable to overspending in upcoming months. Secondary contributors such as expense month-over-month change, health and fitness spending trends, category consistency, and food and dining expenditure reflect the behavioral volatility and category concentration signals that motivated the feature engineering framework. Consistent with Wasserbacher and Spindler, all SHAP-derived feature contributions are interpreted strictly as statistical associations within the forecasting structure, not as causal intervention targets. The model identifies that elevated health and fitness spending is statistically associated with higher overspending risk in the training distribution but it does not assert that reducing such spending will prevent overspending. This distinction is essential for the responsible deployment of XAI in personal finance applications.

6 Limitations

While the results demonstrate the viability of a behavioral forecasting framework for individual overspending prediction, several limitations must be acknowledged.

6.1 Synthetic Dataset

The Sparkov dataset is synthetically generated which means that the spending patterns and class distributions may not fully reflect the complexity and irregularity of real-world consumer financial behavior. Model performance on live transaction data from actual users may differ substantially especially in edge cases that involve irregular income sources, cash transactions, or atypical spending categories.

6.2 Temporal Distribution Shift

As discussed in Section 5.3, the system exhibits significant performance degradation when the class distribution of overspending cases shifts across time periods. The fold 4 drop in F1 from 0.84-0.90 to 0.47 highlights that the models are non-stationary in financial behavior. This limits the reliability of the system in periods of macroeconomic change or seasonal variation without continuous retraining.

6.3 Feature Engineering Ceiling

The study revealed that engineered behavioral features provide only marginal predictive gains over raw monthly spending features. This suggests that the current feature set may not fully capture the behavioral complexity that distinguishes chronic overspenders from situational ones, and that richer longitudinal signals or external contextual data may be required.

7 Future Work

Several directions present meaningful opportunities to extend and strengthen this framework.

7.1 Real-World Transaction Data

Validating the framework on real consumer banking datasets such as those available through open banking APIs or institutional research partnerships would substantially strengthen the generalizability of the findings. Real data introduces noise and behavioral irregularities that synthetic datasets cannot fully replicate.

7.2 Conversational AI Integration

As established by Kamra et al., conversational interfaces can translate complex financial model outputs into plain-language user guidance. Integrating a natural language explanation layer would be capable of contextualizing SHAP contributions for non-technical users and would comp

7.3 Multi-Output and Probabilistic Forecasting

The current system produces a binary overspend/no-overspend prediction. Extending the framework to produce calibrated probability estimates, category-level risk scores, or forecasted overspend magnitude would offer users more nuanced and actionable insight.

8 Conclusion

This paper presented a behavioral forecasting framework for individual-level overspending prediction, combining time-aware feature engineering, supervised machine learning, and explainable AI decomposition via TreeSHAP. Trained and evaluated on over 1.8 million synthetic credit card transactions from the Sparkov dataset and three classifiers which are Logistic Regression, Random Forest, and Gradient Boosting, achieved a strong walk-forward ROC-AUC scores between 0.894 to 0.896 as well as F1 scores up to 0.767 under stable distributional conditions.

The key contributions of this work are threefold. First, the project demonstrates that transaction-derived behavioral indicators can reliably predict next-month overspending risk at the individual level, with income and savings structure emerging as the dominant predictive signals. Second, the feature engineering establishes that behavioral complexity features provide incremental value over raw monthly spending aggregates, particularly for tree-based models. Third, TreeSHAP integration enables transparent decomposition of model predictions into interpretable behavioral contributions which positions the system as a forecasting and transparency tool rather than a causal planning mechanism.

The most significant challenge identified is sensitivity to temporal distribution shift, which substantially degrades performance when overspending prevalence in the test period diverges from training-period distributions. Addressing this through adaptive re-training and drift detection represents the most pressing priority for future development.

Ultimately, this project advances the case that explainable machine learning can move personal financial risk tools beyond static budget thresholds and institutional credit models toward dynamic, behavior-driven, and user-centered early warning systems which makes financial transparency more accessible to everyday individuals.

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